Lecture 2: Binomial Asset Pricing Model

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1 One-Period Binomial Model & Delta Hedge

In an one-period binomial market with risk-free rate r, evaluation of any derivative security can be done by replicating the payoff of the derivative at maturity (time one) using a portfolio of cash and stocks. (We have discussed this last time) On the other hand, one can use delta hedge to price a derivative security.

More specifically, let us denote by $V_1(H)$ and $V_1(T)$ the two possible payoff of a derivative at time one. It is risky, so is the underlying stock S_1. Consider a portfolio consisting of one derivative and $-\Delta_0$ shares of stock, so that the value of the portfolio at time one is risk-free. That is, find a Δ_0 such that

$$V_1(H) - \Delta_0 S_1(H) = V_1(T) - \Delta_0 S_1(T), \tag{1}$$

which yields

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}, \tag{2}$$

the same delta! In an arbitrage-free market, there is only one risk-free rate (try to figure out why). Thus, a risk-free portfolio must grows in value at rate r, which implies that

$$V_0 - \Delta_0 S_0 = \frac{1}{1+r}[V_1(H) - \Delta_0 S_1(H)] = \frac{1}{1+r}[V_1(H) - \Delta_0 S_1(T)], \tag{3}$$

where V_0 is the fair price of this derivative, and Δ_0 is given in (2). Not surprisingly, this will give the same price as replication argument.

2 Risk-Neutral Measure and Multi-Period Binomial Model

As a by-product in solving replication problem, we find the so-called risk-neutral measure. This auxiliary measure highly simplifies pricing of derivative securities, even when the dynamics of the stock is more complex.

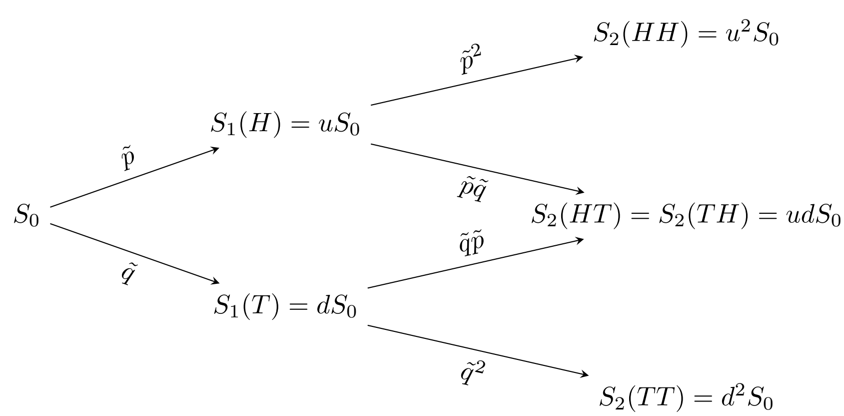

Figure 1: Coin toss and binomial model.

We consider a N-period binomial model in a risk-free world. In Figure 1, $N = 2$, $\tilde{q} = 1 - \tilde{p}$ and

$$\tilde{p} \quad = \quad \frac{1+r-d}{u-d}. \tag{4}$$

In order to price a derivative that matures at time N, we employ the following algorithm

- write down the payoff of this derivative at time two, $V_N(S_N)$ - this is written in the contract; (complexity = number of nodes at level N)
- using backward induction: when $V_{k+1}(S_{k+1})$ is known, $k = N-1, N-2, \ldots, 0$, define

$$V_k(S_k) \quad = \quad \frac{1}{1+r}[\tilde{p}V_{k+1}(uS_k) + \tilde{q}V_{k+1}(dS_k)], \tag{5}$$

$$\Delta_k(S_k) \quad = \quad \frac{V_{k+1}(uS_k) - V_{k+1}(dS_k)}{uS_k - dS_k}. \tag{6}$$

Then $V_k(S_k)$ should be the fair price of the derivative at time k (the value S_k is known at this time). In particular, $V_0 = V_0(S_0)$ is the time-zero fair price of the derivative.

To see why this is true, consider an investor who is trying to replicate the payoff of the derivative (or, if he is trying to hedging the risk of the derivative). He decides to hold a portfolio of $\Delta_k(S_k)$ shares of stock and a cash position of $V_k(S_k) - \Delta_k(S_k) \cdot S_k$ at time k (so that his portfolio has a total value of $X_k(S_k) = V_k(S_k)$). At time $k+1$, his portfolio will have a total value of

$$X_{k+1}(S_{k+1}) \quad = \quad \Delta_k(S_k) \cdot S_{k+1} + (1+r)[V_k(S_k) - \Delta_k(S_k) \cdot S_k]. \tag{7}$$

If the $(k+1)$-th tossing results in a H, then $S_{k+1} = uS_k$,

$$\begin{aligned} X_{k+1}(S_{k+1}) &= \frac{V_{k+1}(uS_k) - V_{k+1}(dS_k)}{uS_k - dS_k} \cdot (u-1-r)S_k + (1+r)V_k(S_k) \\ &= \tilde{q}[V_{k+1}(uS_k) - V_{k+1}(dS_k)] + [\tilde{p}V_{k+1}(uS_k) + \tilde{q}V_{k+1}(dS_k)] \\ &= (\tilde{q}+\tilde{p})V_{k+1}(uS_k) \;=\; V_{k+1}(uS_k). \end{aligned} \tag{8}$$

Similarly, if the $(k+1)$-th tossing results in a T, one can also show that $X_{k+1}(S_{K+1}) = V_{k+1}(dS_k)$. By mathematical (forward) induction, the investor's dynamic portfolio replicates the payoff of the derivative at maturity, $V_N(S_N)$, so no-arbitrage implies that, $V_k(S_k)$ is really the fair price at time k.

Remark 1 *The essence of the above result is that, if one is able to find the risk-neutral measure (by its definition) first, and then pricing is just expectation of the payoff under this measure. This point will not change if we consider more complex model. For example, in Figure 1, if both the up and the down factors u, d depend on time, or even the history of the coin tosses, we can still find the risk-neutral measure first, and then compute the expectation. However, the risk-neutral probability $\tilde{p}$ will now varies with time (and possibly the past coin tosses as well). For details, please refer to the last exercise in Chapter One of the textbook.*

The above algorithm can also be easily **extended** to price a path dependent option. The essential idea is the same. For a rigorous proof, please refer to Theorem 1.2.2 in the textbook.

Example 1. Consider the particular three-period model with $S_0 = 4$, $u = 2$ and $d = \frac{1}{2}$. We have the binomial tree of possible stock price shown in Figure 2. Assume the risk-free rate is $\frac{1}{4}$.[1] Then $\tilde{p} = \tilde{q} = \frac{1}{2}$. Consider a lookback option that pays off

$$V_3 \quad = \quad \max_{0 \le n \le 3} S_n - S_3$$

at time three. Let us denote by $V_k(s, m)$ the price of the option at time k, when $S_k = s$ and $\max_{0 \le n \le k} S_n = m$. Then $V_k(s, m)$ can be determined with the following algorithm:

- when $k = 3$, $V_3(32, 32) = 0$, $V_3(8, 16) = 8$, $V_3(8, 8) = 0$, $V_3(2, 8) = 6$, $V_3(2, 4) = 2$, $V_3(0.5, 4) = 3.5$;

[1] 1 dollar deposited in the money market at time zero will produce an interest of 0.25 dollar at time **one**.

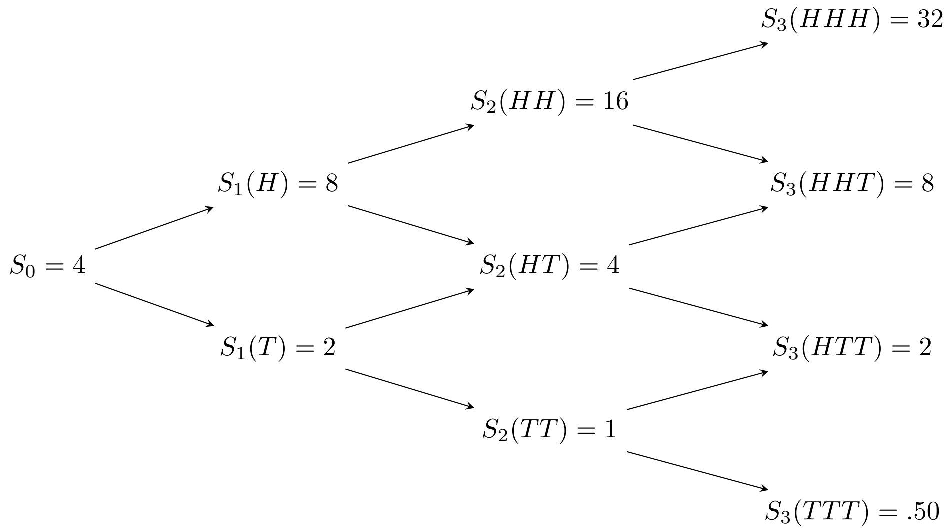

Figure 2: A three-period binomial model

- for $k = 2, 1, 0$, compute

$$V_k(s, m) = \frac{2}{5}[V_{k+1}(2s, \max(2s, m)) + V_{k+1}(\frac{1}{2}s, m)]. \tag{9}$$

The results of the algorithm:

$$\begin{aligned}
V_2(16, 16) &= \tfrac{2}{5}[V_3(32, 32) + V_3(8, 16)] &&= 3.20,\\
V_2(4, 8) &= \tfrac{2}{5}[V_3(8, 8) + V_3(2, 8)] &&= 2.24,\\
V_2(4, 4) &= \tfrac{2}{5}[V_3(8, 8) + V_3(4, 2)] &&= 0.80,\\
V_2(1, 4) &= \tfrac{2}{5}[V_3(2, 4) + V_3(0.5, 4)] &&= 2.20,\\
\\
V_1(8, 8) &= \tfrac{2}{5}[V_2(16, 16) + V_2(4, 8)] &&= 2.24,\\
V_1(2, 4) &= \tfrac{2}{5}[V_2(4, 4) + V_2(1, 4)] &&= 1.20,\\
\\
V_0(4, 4) &= \tfrac{2}{5}[V_1(8, 8) + V_1(2, 4)] &&= 1.376.
\end{aligned}$$

So the time-zero fair price of the option is 1.376. If one wants to replicate the option with stocks and cash, then the number of stocks he should hold $\Delta_k(s, m)$, is given by

$$\Delta_k(s, m) = \frac{V_{k+1}(2s, \max(2s, m)) - V_{k+1}(\frac{1}{2}s, m)}{2s - \frac{1}{2}s}. \tag{10}$$